DS-973 — Data Science & Machine Learning Essentials

PRACTICE EXERCISES, LABS & SOLUTION GUIDE

Course Code: DS-973	**Category:** Computer Science & Tech
Academic Term: 2026 Academic Year	**Format:** E-Learning / Self-Paced
Prerequisites: Basic Problem Solving	**Estimated Hours:** 40 Hours Content

1. Hands-on Practice Exercises & Worksheets

Reinforce your understanding of **Data Science & Machine Learning Essentials** by working through these practical challenges. Attempt each exercise independently before checking the solution guidelines.

Exercise 1: Basic Operations & Validation (Easy)

Problem Statement: Write a function that takes a list of numbers and returns a dictionary containing the count, mean, minimum, and maximum values.

Input Example: [10, 20, 30, 40, 50]
Expected Output: {'count': 5, 'mean': 30.0, 'min': 10, 'max': 50}

Exercise 2: Data Transformation & Filtering (Medium)

Problem Statement: Given a list of student dictionary records, filter out students with a score below 70 and group the remaining student names by course department.

Starter Template:

```
def group_top_students(students: list) -> dict: # TODO: Implement filtering and grouping
logic grouped = {} for s in students: if s.get("score", 0) >= 70: dept =
s.get("department", "General") grouped.setdefault(dept, []).append(s["name"]) return
grouped
```

Exercise 3: Advanced System Integration Challenge (Hard)

Problem Statement: Design an in-memory caching system with key expiration (TTL) and LRU (Least Recently Used) eviction policy.

2. Hints & Solutions Guide

• **Hint for Ex 1:** Use built-in functions len(), sum(), min(), and max().

• **Hint for Ex 2:** Use dict.setdefault() or collections.defaultdict(list) for clean grouping.

• **Hint for Ex 3:** Combine a Hash Map for O(1) lookups with a Doubly Linked List for tracking usage order.